Lesson Planning

Complete Lesson Plan Book

Class: FSc Pre-Engineering · Class 11

Subject: Chemistry

Medium: English

Academic Year 2026-27

DM

This book is prepared as a teaching guide for teachers.

Table of Contents

Unit 1: Chemical Bonding: Ionic and Covalent Bonds

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Learning Objectives

• Students will explain why atoms form chemical bonds in terms of stability.

• Students will distinguish between ionic and covalent bonding with examples.

• Students will predict the bond type in a simple compound from the elements involved.

Key Concepts

• Atoms bond so that they can reach a stable electron arrangement, usually like a noble gas.

• An ionic bond forms when one atom gives electrons to another, making positive and negative ions that attract each other; sodium chloride is a classic example.

• A covalent bond forms when two atoms share pairs of electrons; water and methane are common examples.

• Ionic compounds are usually hard, crystalline solids with high melting points, while covalent compounds are often gases, liquids or soft solids.

• The difference in electronegativity between two atoms is a useful guide to whether their bond is mainly ionic or covalent.

Teaching Methodology (Step by Step)

1. Ask why noble gases rarely react, and use the answers to introduce the idea of a stable electron arrangement.

2. Draw sodium giving one electron to chlorine, and show how the opposite charges hold the ions together.

3. Draw two hydrogen atoms sharing electrons with oxygen to make water, and contrast sharing with transfer.

4. Compare the properties of table salt and wax on the board, and link each property to its bond type.

5. Give five pairs of elements and let students predict the bond type with a reason.

Classroom Activities

• Dot-and-cross diagrams: students draw the electron transfer in sodium chloride and the sharing in a hydrogen molecule.

• Property sort: cards naming properties are sorted into ionic and covalent columns.

• Prediction round: small groups predict the bond type in magnesium oxide, carbon dioxide and potassium bromide, then defend their answers.

Assessment Questions

1. Explain in two lines why atoms form chemical bonds.

2. Sodium chloride is ionic but methane is covalent. Give one reason for each.

3. Predict the bond type in calcium fluoride and justify your answer.

Homework

• Draw dot-and-cross diagrams for magnesium oxide and for a chlorine molecule.

• List three ionic and three covalent substances found at home and give one property of each.

Time Allocation

45 minutes